Status	Finished
Started	Saturday, 17 October 2026, 5:38 PM
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Duration	4 mins 47 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

b. Binary search algorithm is not efficient when the data elements more than 1500 ✓

c. There must be a mechanism to access middle element directly

d. Must use a sorted array

Question **2**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓

b. Complexity

c. Selection

d. Insertion

Question **3**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n

b. n ✓

c. n/2

d. 1

Question **4**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)

b. Hash search

c. Binary search

d. Linear search ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements

b. When the element is found in the list ✓

c. When the search algorithm finishes

d. When the list is sorted

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist

b. The list must be sorted

c. There must be a mechanism to delete and/or insert elements in the list ✓

d. Number values should only be present

Number values should only be present

Question **7**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓

b. O(1)

c. O(log n)

d. O(n log n)

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list is very small
c. When the list is sorted ✓
d. When the list contains duplicate elements

Question **9**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search
b. Linear search
c. Hash search
d. Binary search ✓

Question **10**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search
b. Interpolation search
c. Binary search
d. Linear search ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.
b. Linear search divides the list into halves.
c. Linear search is more efficient than binary search on large lists.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **12**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 1000
c. 500
d. 10 ✓

Question **13**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Binary ✓
b. Hash
c. Both (1) & (3)
d. Linear

Question **14**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all
b. When the item is the last element in the array
c. When the item is somewhere in the middle of the array ✓
d. Item is the last element in the array or item is not there at all

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Linear search ✓
b. Quick search
c. Bubble search
d. Merge search